Acinetobacter Infectious Disease (MONDO:0006635): A Comprehensive Disease Characterization

Summary

Acinetobacter infectious disease is an opportunistic, predominantly healthcare-associated bacterial infection caused chiefly by *Acinetobacter baumannii*, a non-fermenting, aerobic, Gram-negative coccobacillus. It is **not a heritable human disease**; it has no causal human gene, no Mendelian inheritance, and no germline/somatic variant basis. Instead, "disease characteristics" here describe an infectious syndrome whose genetics reside in the *pathogen's* genome (antibiotic-resistance and virulence loci) and whose "risk factors" are clinical and environmental exposures. The organism is one of the six **ESKAPE** pathogens and a **WHO critical-priority** target for new antibiotics. It strikes critically ill, ventilated, and immunocompromised patients—typically in intensive care units—causing ventilator-associated/hospital-acquired pneumonia (VAP/HAP), bloodstream infection (BSI)/sepsis, wound and skin/soft-tissue infection, urinary tract infection, and, less commonly, meningitis and endocarditis.

The clinical danger of *A. baumannii* arises from the convergence of three properties: (1) **extensive drug resistance**, dominated by carbapenem resistance via OXA-type carbapenemases (intrinsic $bla_{\text{OXA-51}}$ plus acquired $bla_{\text{OXA-23}}$) and last-line colistin resistance via PmrCAB-mediated lipid A modification; (2) **exceptional environmental persistence** through resistance to desiccation and disinfectants, enabling survival on surfaces and medical devices and driving nosocomial transmission; and (3) **virulence effectors**, most notably outer membrane protein A (OmpA), which drives host-cell apoptosis via mitochondrial and nuclear targeting, alongside capsule, biofilm, motility, and serum resistance. Together these yield high mortality: in carbapenem-resistant *A. baumannii* (CRAB) nosocomial pneumonia, development of bacteremia raised in-hospital mortality from 45.9% to 68.3%.

Management centers on **combination therapy with ≥2 active agents**, anchored by **sulbactam–durlobactam** (prioritized in IDSA guidance) or **cefiderocol**, plus source control—although high-quality randomized evidence remains limited. Prevention relies on **infection**

prevention and control (IPC) bundles—hand hygiene, contact precautions, environmental cleaning/disinfection, active surveillance, and antimicrobial stewardship—because the hospital environment is the primary CRAB reservoir and **no licensed human vaccine exists**. Experimental therapeutics under investigation include lytic bacteriophages, phage–antibiotic synergy, and anti-biofilm strategies.

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1. Disease Information

Overview. Acinetobacter infectious disease is an opportunistic bacterial infection caused predominantly by *A. baumannii* (and closely related members of the *A. calcoaceticus–A. baumannii* [ACB] complex, including *A. nosocomialis* and *A. pittii*). It is a leading cause of healthcare-associated infection worldwide and a paradigm multidrug-resistant (MDR) organism.

Key identifiers. - **MONDO:** MONDO:0006635 (Acinetobacter infectious disease) - **MeSH:** Acinetobacter Infections (D000151); organism *Acinetobacter baumannii* (NCBI Taxonomy ID 470; genus *Acinetobacter* txid469) - **ICD-10:** A curated code is not disease-specific; *Acinetobacter* is coded via site-specific infection codes plus B96.x ("as the cause of diseases classified elsewhere") - **OMIM / Orphanet:** Not applicable — this is an acquired infectious disease, not a Mendelian/rare genetic disorder - **Category:** Infectious disease

Synonyms / alternative names. Acinetobacter infection; *Acinetobacter baumannii* infection; CRAB (carbapenem-resistant *A. baumannii*) infection; MDR/XDR/PDR *Acinetobacter* infection; historically "*Bacterium anitratum*," "*Herellea vaginicola*," "*Mima polymorpha*."

Data source type. Information is derived from **aggregated disease-level resources** (surveillance datasets, GBD models, clinical cohorts, microbiological studies) rather than individual EHR-linked patient records.

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2. Etiology

Causal factor: infectious (bacterial). The disease is caused by infection with *Acinetobacter* spp., chiefly *A. baumannii*. There is **no human genetic causation**; host genetic risk loci are not established for this infection.

Risk factors (clinical/environmental). A review of severe *A. baumannii* infection identifies the principal risk factors for MDR acquisition as **previous antibiotic use, mechanical ventilation, length of ICU/hospital stay, severity of illness, and use of medical devices** [PMID: 36272902]. Additional cohort evidence flags **advanced age (>75 years)** and **venous catheter use** as independent risk factors for cross-infection in COVID-19 ICU patients, where *A. baumannii* was the dominant organism (53.9% of strains) [PMID: 40624469].

> *"The most common risk factors for the acquisition of MDR A. baumannii are previous antibiotic use, mechanical ventilation, length of ICU and hospital stay, severity of illness, and use of medical devices."* — [PMID: 36272902]

Prognostic risk factor: bacteremia. In a multicenter ICU study of CRAB nosocomial pneumonia, the development of pneumogenic bacteremia was an **independent mortality risk factor (adjusted HR = 2.399, P < 0.001)** [PMID: 38852767].

Protective factors. No genetic protective factors are defined (non-genetic disease). Protective factors are operational/environmental: effective **infection prevention and control**, **antimicrobial stewardship**, and avoidance/limitation of invasive devices. There is no established dietary or germline-variant protection.

Gene–environment interactions. Not applicable in the human-host sense. The relevant "gene–environment" interplay is at the **pathogen** level: antibiotic-selection pressure (environment) selects for resistance genes (OXA carbapenemases, *pmrCAB* mutations), and the built hospital environment selects for desiccation/disinfectant-tolerant clones.

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3. Phenotypes (Clinical Manifestations)

Because this is an infectious disease, "phenotypes" are clinical syndromes and their signs/laboratory abnormalities rather than HPO-encoded heritable traits. HPO terms below are suggested as approximations for knowledge-base mapping.

Syndrome / phenotype	Type	Onset / course	Frequency	Suggested HPO
Ventilator-associated / hospital-acquired pneumonia	Clinical sign + imaging	Acute, healthcare-onset	Most common syndrome; ~93% of ICU respiratory MDR-AB cases developed VAP [PMID: 36641839]	HP:0002090 (Pneumonia)
Bloodstream infection / sepsis / septic shock	Clinical + lab	Acute	Common; raises mortality markedly	HP:0100806 (Sepsis)
Wound / skin & soft-tissue infection	Clinical sign	Acute	Common, esp. trauma/ burns	HP:0100658 (Cellulitis)
Urinary tract infection	Clinical + lab	Acute	Common (catheter-associated)	HP:0000010 (UTI)
Meningitis (post-neurosurgical)	Clinical + CSF	Acute	Uncommon	HP:0001287 (Meningitis)
Endocarditis (e.g., prosthetic-valve)	Clinical + imaging	Subacute	Rare	HP:0001647 (Endocarditis)
Fever	Symptom	Acute	Very frequent	HP:0001945 (Fever)

A. baumannii causes a **spectrum of infections**: VAP/HAP, BSI/sepsis, wound/SSTI, UTI, and less commonly meningitis and endocarditis. A documented CRAB **prosthetic-valve endocarditis** case presented with "*fever, persistent bloodstream infection, cerebellar hemorrhage, and aortic valve vegetation*" [PMID: 42293986]. **Community-onset *A. baumannii* (COAB)** clones with high virulence also exist; 32 Taiwanese COAB isolates were characterized for capsular type, biofilm, motility, desiccation and serum resistance, with in vivo virulence confirmed in a *Galleria mellonella* model, KL49 being the predominant capsular type [PMID: 41402952].

Severity / quality-of-life impact. Infections are typically **severe** in the ICU, with high case-fatality (see §11) and prolonged hospitalization (median length of stay up to ~74 days in one tigecycline cohort [PMID: 22869481]); survivors of ICU sepsis/VAP frequently experience long-term functional impairment. Disease is **acute** rather than chronic, but biofilm-associated device infections may be persistent/relapsing.

4. Genetic / Molecular Information

Not applicable to the human host. There are **no causal human genes, pathogenic germline/somatic variants, modifier genes, human epigenetic changes, or chromosomal abnormalities** underlying susceptibility to *Acinetobacter* infection in any established sense. This section therefore describes the **pathogen's** clinically decisive genetic determinants.

Resistance genes (pathogen). Carbapenem resistance is driven predominantly by **OXA-type carbapenemases**. In 36 clinical isolates from Baghdad, PCR revealed $bla_{\text{OXA-51}}$ in 100% (intrinsic species marker), $bla_{\text{OXA-23}}$ in 86.1%, $bla_{\text{OXA-24}}$ in 69.4%, and $bla_{\text{OXA-58}}$ in 47.2%; carbapenem resistance reached 83.3% (imipenem) and 72.2% (meropenem) [PMID: 42675958].

> *"PCR screening revealed prevalence rates of 100% for bla_{OXA-51}, 86.1% for bla_{OXA-23}, 69.4% for bla_{OXA-24}."* — [PMID: 42675958]

Worldwide, "*OXA-type carbapenemases are the predominant mechanism of resistance; other mechanisms, including efflux pumps, porin mutations, and, rarely, metallo-β-lactamases, may contribute*" [PMID: 42549848]. **Colistin resistance** arises via chromosomal two-component-system mutations (*pmrCAB*) driving lipid A modification (see §6); plasmid *mcr-1–5* genes are **rare/absent** in *A. baumannii* [PMID: 40778526]. Horizontal gene transfer, including **outer membrane vesicle (OMV)-mediated transfer** of resistance determinants, contributes to dissemination.

5. Environmental Information

Environmental factors. *A. baumannii* survives on inanimate hospital surfaces, medical devices, and in moist niches. Its persistence is enabled by **exceptional resistance to disinfectants and desiccation**; compatible solutes contribute to desiccation resistance (studied in *A. baumannii* ATCC 19606) [PMID: 30277310].

> *"Acinetobacter baumannii is a nosocomial pathogen which can persist in the hospital environment not only due to the acquirement of multiple antibiotic resistances, but also because of its exceptional resistance against disinfectants and desiccation."* — [PMID: 30277310]

Lifestyle factors. Not directly applicable; the dominant "exposures" are healthcare-related (ICU admission, ventilation, devices, prior antibiotics). Community-acquired disease is associated in the literature with predisposing conditions (e.g., diabetes, alcohol use, chronic lung disease) in tropical settings, and COAB virulent clones are emerging [PMID: 41402952].

Infectious agent. *Acinetobacter baumannii* (NCBI:txid470), genus *Acinetobacter* (txid469); ACB-complex relatives *A. nosocomialis*, *A. pittii*, *A. calcoaceticus*. CHEBI-relevant chemical entities in pathogenesis include **lipopolysaccharide/lipid A** (CHEBI:16412), **phosphoethanolamine** (CHEBI:58190), and **4-amino-4-deoxy-L-arabinose**.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating exposure → clinical manifestation)

1. **Colonization/exposure:** A critically ill, ventilated, or device-bearing patient is exposed to *A. baumannii* from a **desiccation/disinfectant-resistant environmental reservoir** (surfaces, devices) → **leads to** mucosal/wound colonization. *(Demonstrated; WGS ties clinical isolates to environmental reservoirs — [PMID: 34412693].)*
2. **Adhesion & biofilm:** Surface adhesins, capsule, pili, and **biofilm formation** on host tissue and devices → **result in** protected bacterial communities with reduced drug penetration and immune evasion. *(Demonstrated in vitro — [PMID: 42754003].)*
3. **Invasion & host-cell injury: OmpA** binds host epithelial cells; its N-terminal β-barrel mediates **mitochondrial invasion** → **triggers** a caspase cascade and mitochondrial damage; the C-terminus → **causes** DNA degradation; OmpA also **translocates to the nucleus** → **induces** apoptosis of epithelial cells. *(Demonstrated in vitro/in vivo — [PMID: 41260154]; [PMID: 17760880].)*
4. **OMV branch:** *A. baumannii* releases **OMVs containing OmpA** → **induce** mitochondrial fragmentation and deliver virulence/resistance cargo. *(Demonstrated — [PMID: 33436835].)*

5. **Inflammation & tissue damage:** Host innate recognition of LPS/OmpA → **drives** neutrophilic inflammation, cytokine release, and, in the lung, alveolar epithelial injury → **results in** pneumonia. In an OmpA-deficiency rat model, lung inflammatory damage was alleviated via downregulation of Beclin-1, JNK1, Drp-1, caspase-3/9 and upregulation of Bcl-2 [PMID: 41260154].
6. **Dissemination:** Breach of the epithelial/endothelial barrier → **leads to** bloodstream invasion (bacteremia) → **results in** sepsis/septic shock and secondary seeding (endocarditis, meningitis). Bacteremia independently raises mortality (aHR 2.399) [PMID: 38852767].
7. **Therapeutic failure branch:** Under antibiotic pressure, **OXA carbapenemases** (intrinsic $bla_{\text{OXA-51}}$ + acquired $bla_{\text{OXA-23}}$) → **hydrolyze** carbapenems; **PmrCAB-driven lipid A modification** → **confers** colistin resistance → **results in** failure of last-line therapy and higher death. *(Demonstrated — [PMID: 42675958]; [PMID: 41890236].)*

Detail by category

- **Molecular pathways / cellular processes:** Host **intrinsic (mitochondrial) apoptosis** (caspase-9/-3), **JNK signaling**, and **autophagy modulation** (Beclin-1, Drp-1-mediated mitochondrial fission) are engaged by OmpA [PMID: 41260154]. Relevant GO processes: **apoptotic process (GO:0006915)**, **mitochondrial fragmentation (GO:0090149)**, **inflammatory response (GO:0006954)**, **biofilm formation (GO:0042710)**.
- **Protein dysfunction (pathogen effectors):** OmpA (the most abundant surface protein) is the central virulence effector; its **nuclear translocation via a monopartite NLS induces cytotoxicity**, and nuclear microinjection into *Xenopus* embryos causes embryonic death [PMID: 17760880].
- **Metabolic changes (pathogen):** Integrated proteomics/metabolomics of an XDR-CRAB isolate showed **lipid A modification** and **membrane phospholipid remodeling** underpinning cationic antimicrobial peptide (CAMP)/colistin resistance, with upregulation of PmrA (4.2-fold), PmrB (2.9-fold), and PmrC (5.5-fold) [PMID: 41890236].
- **Immune involvement:** Innate LPS/TLR4 recognition drives inflammation; protective adaptive immunity in models is **Th1/Th17 + humoral** [PMID: 40726064].
- **Biofilm-driven tolerance:** restricted drug penetration, persister cells, efflux activity, quorum sensing and c-di-GMP signaling [PMID: 42754003].

Suggested cell types (CL): alveolar epithelial cells (**CL:0000066** epithelial cell), pulmonary macrophages (**CL:0000235**), neutrophils (**CL:0000775**), vascular endothelial cells (**CL:0000115**).

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7. Anatomical Structures Affected

- **Primary organ: Lung** (UBERON:0002048) — the leading site (VAP/HAP), with alveolar epithelium (UBERON:0002299) and lower respiratory tract injury.
- **Secondary/other sites: Blood/bloodstream** (UBERON:0000178) in bacteremia/sepsis; **skin & soft tissue** (UBERON:0002097) in wound infection; **urinary tract/bladder** (UBERON:0001255) in UTI; **meninges** (UBERON:0002360) in meningitis; **heart valves/endocardium** (UBERON:0002165) in endocarditis; **brain** secondary hemorrhage documented in a CRAB endocarditis case [PMID: 42293986].
- **Body systems:** respiratory, cardiovascular/hematologic, integumentary, urinary, nervous.
- **Tissue/cell level:** epithelial cells (target of OmpA), macrophages and neutrophils (inflammation), endothelium (dissemination).
- **Subcellular compartments (GO CC): mitochondrion (GO:0005739)** and **nucleus (GO:0005634)** are the intracellular targets of OmpA; bacterial **outer membrane (GO:0019867)** and **outer membrane vesicles** are key structures.
- **Lateralization:** pneumonia may be unilateral or bilateral; generally not lateralized as a defining feature.

8. Temporal Development

- **Onset: Acute**. Healthcare-associated disease typically develops after ICU admission/ventilation; in one COVID-ICU cohort the mean time to first cross-infection was 8 ± 5 days [PMID: 40624469]. Community-onset acute infection also occurs [PMID: 41402952].
- **Age of onset:** any age but **heaviest burden in the elderly** (see §9/§11); neonatal ICU cases occur.
- **Progression:** Can progress rapidly from localized infection (pneumonia) to **bacteremia → sepsis → septic shock → multiorgan failure**. In bacteremic CRAB VAP with COVID-19, 91.7% developed septic shock and 57.5% died by 14 days [PMID: 37088438].
- **Course pattern:** Predominantly **acute/monophasic**; device- and biofilm-associated infections may **relapse** or persist.
- **Remission:** Treatment-induced clearance is achievable with active antibiotics + source control; spontaneous clearance in colonized but non-infected patients occurs.

- **Critical period for intervention:** Early appropriate combination therapy and source control before/at the onset of bacteremia is the key window (bacteremia independently worsens survival, aHR 2.399) [PMID: 38852767].

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9. Inheritance and Population (Epidemiology)

Inheritance. Not applicable — acquired infectious disease; no Mendelian inheritance, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency in the human-genetic sense.

Epidemiology / burden. - *A. baumannii* is among the **most common Gram-negative nosocomial isolates** and is predominantly carbapenem-resistant. In an Ethiopian tertiary ICU it was the single most common Gram-negative pathogen (**31.5%** of Gram-negative isolates), predominantly carbapenem-resistant, with **80.1%** of isolates multidrug-resistant [PMID: 42219453]. - It is a **top-five bloodstream-infection death-causing pathogen** in China (GBD 2021), after *S. aureus*, *E. coli*, *S. pneumoniae*, and *P. aeruginosa* [PMID: 40264014]. - It is one of **six pathogens identified as the primary generators of AMR burden** in the WHO Eastern Mediterranean Region (with *S. pneumoniae*, *K. pneumoniae*, *E. coli*, *S. aureus*, *P. aeruginosa*) [PMID: 41082884]. - **Bloodstream and lower respiratory infections account for the majority of AMR-related mortality** globally [PMID: 42674241].

Demographics. Burden **increases with age**, heaviest in those **≥70 years** [PMID: 40264014]; associated deaths in adults ≥70 rose by ~85.7% (1990–2021) in the EMR [PMID: 41082884]. Male predominance is reported in respiratory-niche isolates (elderly male patients). Geographic distribution is **global**, with high CRAB prevalence in Asia, the Middle East, Latin America, and southern/eastern Europe; some settings show recent declines (Thailand: 12% decline in hospital-acquired CRAB BSI frequency per 100,000 admissions, 2022→2024) [PMID: 42202898].

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10. Diagnostics

Culture-based identification. Diagnosis is **culture-based**, with species identification by **MALDI-TOF mass spectrometry**. Standard MALDI-TOF can misassign closely related ACB-complex species (*A. baumannii* vs *A. nosocomialis*), requiring optimized protocols (ferulic-acid matrix) for reliable differentiation [PMID: 24054697].

> *"this method suffers from the inability to differentiate reliably between some closely related bacterial species including those of the Acinetobacter calcoaceticus–Acinetobacter baumannii (ACB) complex, namely A. baumannii and Acinetobacter nosocomialis."* — [PMID: 24054697]

Rapid molecular diagnostics. Rapid molecular blood-culture identification panels (e.g., GenMark ePlex BCID-GN) detect *A. baumannii* and deliver results **~26.7 h earlier** than conventional workflows, with high concordance vs MALDI-TOF [PMID: 36688641]. **NGS/tNGS** improves detection of non-culturable and polymicrobial organisms and can predict AMR markers (KPC, NDM, OXA-48, mecA) with moderate accuracy [PMID: 42106793; 39328359].

Resistance-gene / mechanism testing. PCR for $bla_{\text{OXA-51/23/24/58}}$ characterizes carbapenemase content [PMID: 42675958]. Susceptibility testing follows **CLSI** guidelines; **MALDI-based lipid A ("MALDIxin") tests** can rapidly flag colistin resistance and distinguish chromosomal from *mcr*-mediated mechanisms [PMID: 31580426].

Clinical criteria / differential diagnosis. Diagnosis integrates culture positivity with a compatible syndrome (VAP/HAP, BSI, etc.). Differential diagnosis includes other Gram-negative nosocomial pathogens (*P. aeruginosa*, *K. pneumoniae*, *E. coli*) and, in polymicrobial LRTI, co-infection with *P. aeruginosa* is common [PMID: 42106793].

Screening. Active surveillance cultures (respiratory, rectal, environmental) are used for outbreak detection and cohorting, not population screening [PMID: 34412693].

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11. Outcome / Prognosis

Mortality. High. In CRAB nosocomial pneumonia, **bacteremia raised in-hospital mortality from 45.9% to 68.3% ($P < 0.001$)** and was an independent risk factor (aHR = 2.399) [PMID: 38852767]. In bacteremic CRAB VAP in COVID-19 patients, **14-day mortality was 57.5% and 30-day mortality 80.8%**, with septic shock in 91.7% [PMID: 37088438]. A Saudi ICU MDR-AB respiratory cohort reported a **74% death rate** [PMID: 36641839]. In an MDR-AB tigecycline cohort, in-hospital mortality was **51.9%** [PMID: 22869481].

Cohort / setting	Population	Mortality	PMID
Multicenter ICU, CRAB pneumonia	Bacteremic vs non-bacteremic	68.3% vs 45.9%	38852767
COVID-ICU, bacteremic CRAB VAP	73 patients	57.5% (14 d); 80.8% (30 d)	37088438
Saudi ICU, MDR-AB respiratory	115 ventilated	74%	36641839
Tertiary hospital, MDR-AB, tigecycline	27 patients	51.9% in-hospital	22869481

Prognostic factors. Bacteremia, septic shock, higher comorbidity index (Charlson), ICU status, ECMO, inappropriate empiric therapy (lacking colistin/tigecycline), and COVID-19 co-infection worsen outcomes [PMID: 38852767; P 37088438 ; P 36641839 ; P 38634975]. **Morbidity** includes prolonged ventilation, extended ICU/hospital stay, and secondary complications (superinfection, organ failure). **Recovery** is achievable with early active therapy and source control.

12. Treatment

Guideline framework. IDSA (2024/2026) and ESCMID both address CRAB. A 2026 therapeutic review notes both societies support **combination therapy with ≥2 active agents** for severe CRAB despite limited RCT data [PMID: 42549848; P 42570093 ; P 39108079].

> *"provides treatment suggestions for infections caused by ... carbapenem-resistant Acinetobacter baumannii (CRAB) ..."* — IDSA 2026 [PMID: 42570093]

Core / preferred agents. - **Sulbactam–durlobactam** — prioritized as the **core of treatment** in IDSA guidance [PMID: 42549848]. *(NCIT: sulbactam C47707.)* - **Cefiderocol** — a novel siderophore cephalosporin suggested for **refractory/resistant infection or prior treatment failure** [PMID: 42549848]; clinically effective for CRAB BSI with higher clinical cure (66% vs 44.4% for colistin, P = 0.027) and fewer adverse events, and lower mortality/higher cure in bacteremic HAP/VAP [PMID: 38634975]; cefiderocol ± fosfomycin independently associated with 30-day survival in bacteremic CRAB VAP [PMID: 37088438]. - **High-dose ampicillin–sulbactam**, **polymyxins/colistin**, and **tetracyclines (tigecycline, minocycline, eravacycline)** remain combination components [PMID: 42549848]. Tigecycline monotherapy shows high microbiological eradication but lower clinical success and should be limited in critically ill VAP/BSI [PMID: 22869481; P 21789524].

Combination regimens. Combination is the **preferred but evidence-limited** strategy for carbapenem-resistant Gram-negatives [PMID: 34107899]. A salvage regimen of **tigecycline** + **extended-infusion imipenem** outperformed sulbactam+imipenem in XDR-AB VAP bacteremia (mortality 14.3% vs 64.3%), correlating with in vitro synergy [PMID: 26341302].

> *"the combination of antibiotics has become the preferred treatment strategy for CR-GNB infection. However, robust evidence to support this approach is lacking."* — [PMID: 34107899]

Pharmacogenomics. Not applicable (no host-genotype-guided dosing established); colistin nephrotoxicity is a key adverse-event consideration [PMID: 38634975].

Experimental / advanced therapeutics. Lytic **bacteriophages** and **phage–antibiotic synergy** are under active investigation; a novel lytic phage (vB_AbaM_EA1) infected 27.5% of MDR isolates, disrupted mature biofilms, and lacked toxin/lysogeny genes [PMID: 42469629; P 42792069].

Phage efficacy can be undermined by **anti-phage antibodies** (fiber/nozzle-specific) [PMID: 42771684]. **Anti-biofilm strategies** (quorum-sensing inhibitors, matrix-degrading enzymes, nanoparticles, antimicrobial peptides) target biofilm-driven tolerance [PMID: 42754003].

Suggested NCIT terms: Antibiotic Therapy (C15370), Combination Chemotherapy, Cefiderocol, Sulbactam, Colistin, Tigecycline, Bacteriophage Therapy.

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13. Prevention

No human vaccine exists. Prevention centers on **infection prevention and control (IPC)** and antimicrobial stewardship.

IPC bundles. Core interventions are **hand hygiene, targeted contact precautions, optimized environmental cleaning, and antimicrobial stewardship**, with active surveillance and cohorting in outbreaks [PMID: 42055841]. A **five-component IPC bundle** (hand hygiene, screening, universal contact precautions, environmental sampling, one-time radical environmental cleaning/disinfection) reduced ICU-CRAB incidence density **from 30.4 to 0 cases per 1000 patient-days**; WGS confirmed clinical isolates were clonally related to an **environmental reservoir**, establishing the environment as the primary route of spread [PMID: 34412693].

> *"nosocomial ICU-CRAB ID decreased from 30.4 CRAB cases per 1000 patients-days to zero cases per 1000 patients-days."* — [PMID: 34412693]

Outbreak control. Switching disinfectants to **sodium hypochlorite/phenolic compounds**, enhanced hand hygiene and PPE halted a COVID-ward CRAB outbreak; WGS showed clonal identity between patient and environmental isolates [PMID: 38686643; P 39335042].

Immunization. Experimental only — OmpW-based multiepitope nanovaccines and OMV vaccines protect mice against pneumonia/bacteremia (see §15) [PMID: 40726064; P 38773507].

Prevention levels: primary = IPC + stewardship; secondary = surveillance/early detection; tertiary = prompt combination therapy + source control to prevent bacteremia/complications.

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14. Other Species / Natural Disease

- **Taxonomy of pathogen:** *Acinetobacter baumannii* (NCBI:txid470); genus *Acinetobacter* (txid469).
- **Host range / models:** Disease is modeled in **mice** (NCBI:txid10090; acute lethal pneumonia and bacteremia) and the invertebrate ***Galleria mellonella*** (greater wax moth) larva as a virulence model [PMID: 40726064; P 38773507 ; P 41402952].
- **Veterinary / natural disease:** *Acinetobacter* spp. are widespread environmental organisms and opportunistic pathogens; specific naturally occurring companion-animal disease was not a focus of this investigation (OMIA not queried; data gap).
- **Zoonotic potential:** *A. baumannii* is primarily environmental/nosocomial rather than a classical zoonosis; livestock/companion-animal reservoirs of MDR clones are an area of One-Health interest (not directly assessed here).

15. Model Organisms

- **Murine models:** acute lethal **pneumonia** and **bacteremia** models are standard for virulence and vaccine testing [PMID: 40726064; P 38773507].
- **Invertebrate model:** ***Galleria mellonella*** larvae for rapid in vivo virulence assessment [PMID: 41402952].
- **Rat model:** SD rats used to show OmpA-dependent lung inflammatory damage (OmpA deficiency alleviated injury via Beclin-1/JNK1/Drp-1/caspase-3/9 down- and Bcl-2 up-regulation) [PMID: 41260154].
- **Xenopus embryo:** used to demonstrate OmpA nuclear cytotoxicity (nuclear microinjection causes embryonic death) [PMID: 17760880].
- **Vaccine models:** OmpW-based dendritic-cell-targeting **multiepitope nanovaccine** (Th1/Th17 + humoral, protective in pneumonia) [PMID: 40726064]; **OMV vaccines** (LOMV/nOMV)

protective in pneumonia and bacteremia [PMID: 38773507]; heat-killed/heat-inactivated mucosal vaccination approaches [PMID: 41058705].

- **Phenotype recapitulation:** murine pneumonia/bacteremia models recapitulate acute lethal infection, host inflammation, and permit protective-immunity and phage-therapy testing; **limitations** include species differences in innate immunity and inability to fully model chronic device/biofilm infection and human ICU comorbidity.

Mechanistic Model / Interpretation

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ENVIRONMENTAL RESERVOIR (desiccation/disinfectant-resistant)   [PMID:30277310]
       │  survives on surfaces & devices
       ▼
COLONIZATION of ventilated/device-bearing ICU patient          [PMID:36272902]
       │  adhesins, capsule, pili
       ▼
BIOFILM on tissue/devices → drug tolerance, immune evasion      [PMID:42754003]
       │
       ▼
HOST-CELL INVASION & INJURY via OmpA
  ├─ N-terminus → mitochondrial damage → caspase cascade        [PMID:41260154]
  ├─ C-terminus → DNA degradation → apoptosis
  ├─ nuclear translocation (NLS) → cytotoxicity                 [PMID:17760880]
  └─ OMV-delivered OmpA → mitochondrial fragmentation           [PMID:33436835]
       │
       ▼
INFLAMMATION & ALVEOLAR EPITHELIAL INJURY → PNEUMONIA (VAP/HAP) [PMID:36641839]
       │  barrier breach
       ▼
BACTEREMIA → SEPSIS/SEPTIC SHOCK (mortality ↑, aHR 2.399)       [PMID:38852767]
       │
       ▼
THERAPEUTIC FAILURE BRANCH
  ├─ OXA carbapenemases (OXA-51 intrinsic + OXA-23 acquired)    [PMID:42675958]
  └─ PmrCAB lipid A modification → colistin resistance          [PMID:41890236]
       │
       ▼
HIGH MORTALITY (up to ~68–81% in bacteremic CRAB)              [PMID:37088438]
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The pathophysiology is best understood as a **pathogen-driven, host-injuring cascade** in which environmental hardiness and drug resistance determine *whether and how severely* infection occurs, while OmpA-centered virulence determines *how host cells are killed*. Upstream levers (environment, devices, antibiotic pressure) are the actionable prevention targets; downstream events (bacteremia, resistance) are the prognostic determinants.

Evidence Base

PMID	Contribution	Finding
42219453	Leading Gram-negative ICU pathogen (31.5%), 80.1% MDR	F001
42178970	ESKAPE priority classification	F001
42675958	OXA carbapenemase gene prevalence	F002
42549848	OXA predominant mechanism; sulbactam-durlobactam/cefiderocol	F002, F004
36272902	Risk factors for MDR acquisition	F003
38852767	Bacteremia independent mortality risk (aHR 2.399)	F003
42570093	IDSA 2026 CRAB guidance	F004
41402952	Community-onset virulent clones; Galleria model	F005
42293986	CRAB prosthetic-valve endocarditis	F005
30277310	Desiccation/disinfectant resistance	F006
40264014	Top-5 BSI death pathogen (China)	F007
42674241	BSI/LRTI dominate AMR mortality	F007
40726064	Murine pneumonia model; OmpW vaccine	F008
38773507	OMV vaccine; pneumonia + bacteremia models	F008
41260154	OmpA apoptosis mechanism	F009
17760880	OmpA nuclear cytotoxicity	F009
33436835	OMV-OmpA mitochondrial fragmentation	F009
24054697	MALDI-TOF ACB-complex differentiation	F010
36688641	Rapid molecular BCID panel (~26.7 h faster)	F010
41890236	PmrAB/PmrC lipid A modification	F011
40778526	mcr genes rare/absent	F011
34412693	5-component IPC bundle eliminates CRAB	F012
42055841	Core IPC interventions	F012
42469629	Anti-biofilm lytic phage	F013

PMID	Contribution	Finding
34107899	Combination therapy preferred but evidence-limited	F013

Supporting/context papers: 37088438, 38634975, 26341302, 22869481, 21789524, 36641839, 40624469, 41082884, 42202898, 42106793, 39328359, 31580426, 42754003, 42771684, 38686643, 39335042, 39108079.

Limitations and Knowledge Gaps

1. **Not a genetic human disease.** Sections designed for Mendelian disorders (causal genes, variants, inheritance, penetrance, epigenetics, chromosomal abnormalities, genetic counseling, newborn/carrier screening) are **not applicable**; host susceptibility genetics for *Acinetobacter* infection are not established.
2. **Species heterogeneity.** Most evidence concerns *A. baumannii*; other ACB-complex members (*A. nosocomialis*, *A. pittii*) and non-baumannii *Acinetobacter* differ in virulence and resistance and are under-characterized. MALDI-TOF misidentification may bias epidemiology [PMID: 24054697].
3. **Treatment evidence quality.** Combination-therapy superiority rests largely on observational/quasi-experimental data; **robust RCTs are lacking** [PMID: 34107899]. Newer agents (sulbactam-durlobactam, cefiderocol) have limited head-to-head CRAB-specific outcome trials.
4. **Geographic bias.** Surveillance is concentrated in Asia, the Middle East, and select high-income settings; low-income community-onset burden is under-measured.
5. **Model limitations.** Murine/Galleria models do not capture chronic device biofilm infection, human comorbidity, or ICU pharmacokinetics; no licensed human vaccine has emerged despite promising preclinical candidates.
6. **Veterinary/One-Health data** (OMIA, animal reservoirs, zoonotic transfer) were not systematically investigated — a genuine gap.

Proposed Follow-up Experiments / Actions

1. **Randomized controlled trials** comparing sulbactam–durlobactam- vs cefiderocol-anchored combination regimens for CRAB bacteremia/VAP, powered for 28-day mortality.
2. **Host-susceptibility genomics:** GWAS/immunogenetic studies in ICU cohorts to test whether host innate-immune variants modulate *Acinetobacter* infection risk/outcome (currently absent).
3. **OmpA-targeted therapeutics:** evaluate OmpA-neutralizing antibodies or small molecules blocking mitochondrial/nuclear translocation as adjuncts, building on the OmpA-deficiency rat protection data [PMID: 41260154].
4. **Phage-therapy optimization:** engineer phage cocktails and mitigate anti-phage antibody-mediated failure (fiber/nozzle epitope engineering) demonstrated in mice [PMID: 42771684].
5. **Vaccine advancement:** move OmpW nanovaccine and OMV candidates toward first-in-human immunogenicity/safety studies [PMID: 40726064; P 38773507].
6. **Environmental/One-Health surveillance:** integrate WGS-based environmental sampling (proven to trace reservoirs [PMID: 34412693]) with veterinary/community sampling to map transmission and MDR-clone reservoirs.
7. **Rapid diagnostics rollout:** deploy molecular BCID panels and MALDI lipid A testing to shorten time-to-appropriate therapy and detect colistin resistance early [PMID: 36688641; P 31580426].

Evidence source types represented: human clinical (cohorts, surveillance, guidelines), model organism (mouse, rat, Galleria, Xenopus), in vitro (isolate characterization, proteomics/metabolomics, phage assays), and computational/epidemiological (GBD modeling).
